KalMydas

Whitepaper

Algorithmic Trading Meets Decentralized Technology

Version 1.1 — April 2026

Network: Base (mainnet, planned September 14, 2026) / Arbitrum Sepolia (testnet)

1 Executive Summary

KalMydas is a decentralized algorithmic trading protocol built on Base (mainnet) with testnet on Arbitrum Sepolia that bridges the gap between professional-grade quantitative trading and accessible DeFi result generation. The platform deploys five distinct XAUUSD (Gold) trading algorithms, each targeting a different risk-result profile, and distributes performance transparently on-chain.

Unlike traditional DeFi token emission protocols that rely on token emissions or unsustainable yields, KalMydas generates real results from algorithmic trading on global gold markets. Every trade is recorded on-chain with a unique Trade ID, every fee distribution is verifiable on Arbiscan, and every pool's performance is displayed in real time.

The KAL token (fixed supply: 10,000,000) serves as the governance and utility backbone of the ecosystem. Through the veKAL (vote-escrowed KAL) mechanism inspired by Curve Finance, long-term holders earn boosted LP rewards (up to 2.5x), weekly USDC fee sharing, and governance voting power over protocol parameters.

Core Principles

- Real Performance, Not Emissions — Results come from trading performance, not inflationary rewards.
- Radical Transparency — Every trade, fee, and distribution is verifiable on-chain.
- Zero Burn Philosophy — No tokens are destroyed. All value recirculates as Protocol-Owned Liquidity (POL).
- Resilience by Design — Stress-tested across 12 extreme scenarios over 100-year simulations.
- Progressive Decentralization — From founding team operations to full community governance via veKAL voting.

2 The Problem

The majority of DeFi protocols generate results via token emissions, creating a circular dependency. When emissions decrease or market sentiment shifts, capital flight creates death spirals. DeFi needs protocols that generate returns from external sources.

Professional algorithmic trading has historically been the domain of hedge funds and institutions. KalMydas democratizes access to institutional-grade Gold trading algorithms.

Even in DeFi, many protocols operate as black boxes. Users have no visibility into how results are generated. KalMydas solves this with total on-chain transparency.

3 The KalMydas Solution

KalMydas bridges centralized algorithmic trading with decentralized finance infrastructure. The platform operates five distinct Gold (XAUUSD) trading algorithms on MetaTrader 4. Trade results are reported on-chain daily by a permissioned Oracle.

The 5-Step Flow

- Access — Acquire an Access Pass (subscription Soulbound NFT) to unlock pool deposits. 4 tiers: Bronze, Silver, Gold, Diamond.
- Deposit — Deposit USDC into one or more of the five strategy pools.

• Trading — Algorithms trade XAUUSD 24/5. A Bot Operator reports daily P&L to smart contracts with unique Trade IDs.

• Distribution — Fees automatically distributed: 35% reserve, 20% POL, 15% LP, 10% veKAL fee sharing, 10% founder, 5% pool, 5% RWA.

• Compound or Withdraw — Withdraw anytime or use the Auto-Vault (ERC-4626) to auto-compound.

4 Trading Strategies

Five XAUUSD (Gold) Algorithms

Algorithm	Profile	TF	Direction	PF	DD Max	Win Rate	Trades	CAGR	Fees	KAL Bonus
HORIZON	Conservative	H4	Buy Only	1.31	-46.01%	21.49%	1,098	4.15%	~1.2%/10%	+5%
VALKYRIE	Balanced	D1	Bidirectional	2.30	-59.14%	46.81%	329	12.49%	~2.4%/20%	+3%
REVOLUTION	Dynamic	H1	Bidirectional	1.11	-63.66%	35.00%	2,166	24.43%	~6%/20%	+2%
TREASURY	Aggressive	H1	Bidirectional	1.35	-70.26%	38.18%	1,815	33.05%	~6%/20%	+2%
ORION	High-Risk	H4	Bidirectional	1.65	-31.36%	64.88%	800	78.99%	0%/20%	+2%

5 KAL Token Economics

The KAL token is an ERC-20 utility token with a hard-capped supply of 10,000,000 tokens. The emission model follows a Bitcoin-inspired halving schedule designed to transition from growth incentives to a self-sustaining protocol economy.

Conditional Presale (Pattern A — 10 Tiers)

Schema validated at session 277: presale between July 15 and August 31, 2026, with a viability gate set at ≥ 75,000 USDC (Phase 1 sold out — tiers 0+1+2 = 50K × ($0.30 + $0.50 + $0.70)). Audited contract hard cap is 482,500 KAL + 17,500 KAL bonus = 500,000 KAL (5% of total supply). Maximum raise: 1,510,000 USDC. If the viability gate is met, mainnet deployment on Base on September 14, 2026; otherwise 4 to 8 week delay, with no personal capital engaged.

Tier	Name	Price	Allocation	Bonus
0	SEED	$0.30	50,000 KAL	+20%
1	STRATEGIC	$0.50	50,000 KAL	+10%
2	COMMUNITY	$0.70	50,000 KAL	+5%
3	PUBLIC	$1.50	50,000 KAL	-
4	PUBLIC+	$2.50	50,000 KAL	-
5	GROWTH	$3.50	50,000 KAL	-
6	GROWTH+	$4.50	50,000 KAL	-
7	PREMIUM	$5.50	50,000 KAL	-

Tier	Name	Price	Allocation	Bonus
8	PREMIUM+	$6.00	50,000 KAL	-
9	FINAL	$8.00	32,500 KAL	-

6 veKAL: Vote-Escrowed Governance

Users lock KAL tokens for 1 week to 4 years to receive non-transferable veKAL. Formula: veKAL = KAL locked × (time remaining / 4 years).

Four Utilities

- Governance Voting — 1 veKAL = 1 vote on protocol proposals.
- LP Boost — Up to 2.5x multiplier on mining rewards.
- Fee Sharing — Weekly USDC from performance fees.
- Fee Reduction — Up to -30% on Access Pass fees.
- Governance details: Proposal threshold 1% total veKAL. Quorum 10%. Voting 3 days. Timelock 48h. Grace period 7 days.
- Early locker bonus: Phase 1 (weeks 1-4) 3x, Phase 2 (weeks 5-12) 2x, Phase 3 (weeks 13-26) 1.5x.

7 Fee Architecture

Performance Fee Split (KalFeeSplitV7)

Destination	%	Description
Reserve	35%	Pool safety buffer
Protocol-Owned Liquidity	20%	Permanent market depth
LP Mining Rewards	15%	Bonus KAL to LPs
veKAL Fee Sharing	10%	Weekly USDC to veKAL holders
Founder	10%	Team compensation
Pool Rewards	5%	KAL bonus for depositors
RWA Treasury	5%	Tokenized T-Bills rate

8 Access Pass

The KalAccessPassV5 is a mandatory soulbound (non-transferable) NFT for deposits. KAL payment = 30% discount. Distribution: 70% founder, 20% Buyback, 10% Rainy Day Fund.

9 Referral Program

Qualification: Active Access Pass + min $100 USDC deposit + 30 days active. 4-month linear vesting. Annual budget capped at 2% of total supply.

10 Risk Management

Active Protections

- Rainy Day Fund — USDC reserve (capped at $60,000) funded by 5% of performance fees. Automatic buyback after 2 consecutive negative months.
- Capital Manager — Real-time pool health monitoring (HEALTHY/LOW/HIGH/CRITICAL). Automatic bidirectional rebalancing.
- Result Diversifier — If bot is inactive >7 days, 50% of assets are automatically redirected to Aave/Compound.
- RWA Treasury — 5% of fees into tokenized US T-Bills (Ondo USDY). ~4.5% rate independent of crypto markets.

11 Security & Audit

Independent final retest validated April 22, 2026 by Omniscient Security Labs (PASS — technical blockers cleared), followed on April 25, 2026 by a third consecutive PASS on the KalPool individual pause function (pauseMyStrategy / liftMyPause — read-only preference flag, 0 value transfer, EIP-170 margin of 418 bytes independently confirmed, GO Sepolia redeployment May 3, 2026). 42 smart contracts audited in total: 37 core contracts (Omniscient external + KalMydas Labs internal + peripheral audit) plus the 5 KalSoloPool added at STEP 4.24 (user-choice mode, deployed on Arbitrum Sepolia April 25, 2026). 309,115 interactions tested: 1,115 Hardhat tests + 8,000 Foundry Forge attacks on KalPresaleV2 and VeKAL + 300,000 Foundry Forge transitions on KalSoloPool (6 invariants × 50,000 calls, 0 reverts, 0 violations). 33 findings addressed in total. Audit policy retained for mainnet: Omniscient Labs as sole vendor.

Security Layers

- ReentrancyGuard on all contracts
- SafeERC20 for all transfers
- Emergency Pausable
- 24-48h operator Timelock
- Full anti-MEV protection (minOut + off-chain deadline)
- veKAL temporal checkpoints (anti flash-loan)
- Soulbound LP tokens (anti-bypass)
- Circuit Breaker with O(1) ring buffer
- Temporal voting power decay (Curve model)
- High-Water Mark for fees
- Operator withdrawal limit

12 Roadmap

Phase	Period	Milestones	Status
Phase 1	Q1 2026	VeKAL + FeeDistributor + 1,109 tests	COMPLETE
Phase 2	Q1 2026	Governance, Referral, Auto-Vault, Zero Burn, RWA, Capital Manager	COMPLETE
Phase 3	Q1 2026	20+ pages frontend, Privy wallet, on-chain data, smart router	COMPLETE
Phase 4	Avr. 2026	Internal audit + external Omniscient + 5 KalSoloPool deployed on Sepolia + final retest April 22 PASS + 300,000-t	COMPLETE
Phase 5	15 jul. → 31 août 2026	Pattern A conditional presale — ≥ $75K USDC gate (Phase 1 sold out, audited contract cap), 500K KAL cap (5% s	PLANNED
Phase 6	14 sept. 2026	Base deployment via atomic Gnosis Safe batch (Path 2), conditional on presale gate	CONDITIONAL

Mainnet Security Plan

- Gnosis Safe Multisig 2/3 (planned at launch)
- Timelock connection to emergency functions
- Active community bug bounty
- Progressive decentralization to governance
- Conditional presale schema — viability gate ≥ 75,000 USDC between July 15 and August 31, 2026, mainnet September 14, 2026 on Base if gate is reached, otherwise 4 to 8 week delay
- Alpha + Beta vesting without KalToken modification via atomic Gnosis Safe batch (Path 2)

Disclaimer

This document is for informational purposes only and does not constitute financial advice. The KAL token is a utility token. Participation involves significant risks, including potential loss of all deposited assets. Past performance does not guarantee future results.